Rarefied-Flow CFD and DeepONet Case Studies

Micro-Step and Micro-Nozzle: from DSMC verification to held-out fields

Week 9 Extension Lecture Notes

M&I-ENG 690A – AI in Fluid Mechanics

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Central question. How do we establish credible DSMC labels first, then train and test a neural operator without confusing flow-solver verification, data integrity, and surrogate error?

Instructor: Ehsan Roohi

Contents

1 Position in the Course

Week 9 joins two 2026 Roohi–Mahdavi studies to the course's earlier DSMC, operator-learning, and validation material. The micro-step varies rarefaction or step height; the micro-nozzle varies back pressure and, in the article, the throat location. The numerical solver and the learned operator have separate questions, evidence, and acceptance gates.

Learning checkpoint

By the end of the lecture and two laboratories, a student should be able to:

1. summarize the DSMC move–index–collide–boundary–sample algorithm;
2. design particle, mesh, time-step, sampling, and conservation checks;
3. write the branch–trunk form of DeepONet and a POD-trunk variant;
4. split complete physical cases into development, validation, and held-out groups;
5. interpret global, local-zone, centerline, and full-field errors together;
6. regenerate the FlowMLLab back-pressure figures from public DSMC fields; and
7. state which article results are references and which are new executable results.

1.1 The three evidence layers

Layer	Question	Minimum evidence
DSMC solver	Does the particle calculation approximate the intended kinetic problem?	$\Delta x/\lambda$, $\Delta t/\tau_c$, particles/cell, sampling uncertainty, boundaries, conservation
Dataset	Are the retained arrays the intended solver outputs?	source revision, hashes, units, grid, finite values, case metadata
Neural operator	Does the fitted map generalize to new physical cases?	case-wise split, frozen selection, baselines, local and global errors, failure cases

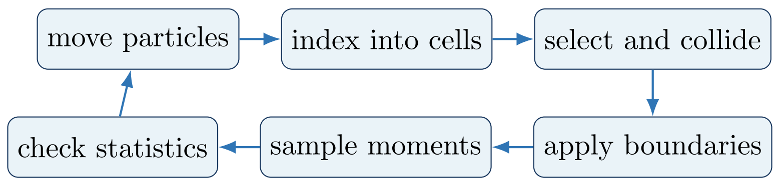
2 DSMC as an Independent Computational Experiment

For a dilute gas, the Knudsen number compares a molecular mean free path to a characteristic length,

$$\mathrm{Kn} = \frac{\lambda}{L}. \tag{1}$$

When continuum closure becomes questionable, DSMC advances representative particles while separating free motion and binary collisions over a short time step.

2.1 One DSMC time step



1. **Move:** update position from molecular velocity during Δt.
2. **Boundary interaction:** process inlet, outlet, symmetry, and wall reflection/accommodation.
3. **Index:** place particles in collision cells or subcells.
4. **Collide:** choose candidate pairs; an NTC scheme accepts pairs according to relative speed and cross section.
5. **Sample:** after a transient, accumulate number, momentum, and energy moments over many statistically correlated steps.

The familiar design rules are local: collision cells should resolve the local mean free path, the time step should resolve the local mean collision time, and particle count and sample duration must make the uncertainty acceptable in the specific observable of interest.

Evidence gate

A credible CFD/DSMC submission reports at least two spatial resolutions, two time steps, particles per cell, sampling length, independent seeds or block statistics, wall model, inlet/outlet treatment, mass balance, and uncertainty on the final comparison quantities. A smooth contour is not a convergence test.

3 From Fields to Operators

A parameter-to-field operator predicts the response at a query coordinate $\boldsymbol{y}$ for a physical input $\boldsymbol{p}$. A standard DeepONet writes

$$\widehat{G}(\boldsymbol{p})(\boldsymbol{y}) = \sum_{k=1}^{r} b_k(\boldsymbol{p}) t_k(\boldsymbol{y}) + b_0. \tag{2}$$

The branch produces coefficients b_k; the trunk produces coordinate features t_k. In a POD–DeepONet, the trunk is a set of POD modes learned from the development fields, while a neural branch predicts their coefficients.

3.1 Case-wise experimental design

For the nozzle back-pressure study, FlowMLLab uses

$$\mathcal{D}_{\text{dev}} = \{15, 18, 19, 20, 22, 23, 24, 26, 27, 28, 29, 33\} \text{ kPa}, \tag{3}$$

$$\mathcal{D}_{\text{test}} = \{16, 25, 30\} \text{ kPa}. \tag{4}$$

Two full-field baselines are compared inside $\mathcal{D}_{\text{dev}}$: physical-coordinate interpolation and interpolation after aligning row-wise density-compression surfaces. Alignment is selected per output only when it reduces mean shock-window error and increases mean global error by at most two percentage points. The held-out fields are evaluated after that rule and all metric definitions are fixed.

For a reference field q and prediction $\widehat{q}$,

$$\varepsilon_{L_2}(q, \widehat{q}) = \frac{\|\widehat{q} - q\|_2}{\|q\|_2} \times 100\%. \tag{5}$$

This metric must be supplemented by localized error, shock position, reverse- flow topology, conservation, and uncertainty of the DSMC reference. FlowMLLab therefore also reports a row-wise window within $\pm 3\delta_j$ of the reference shock and a density-gradient-weighted norm with $w = 1 + 4|\partial\rho/\partial x| / \max|\partial\rho/\partial x|$. The reference shock is used only to score a completed prediction; the predicted aligned coordinate comes solely from bracketing development fields.

Scientific boundary

Model error relative to DSMC is not DSMC error relative to the physical flow. The first evaluates the operator against its numerical labels. The second requires independent solver verification and external validation evidence.

4 Micro-Step: Physics and Zonal Learning

The backward-facing micro-step creates separation, reverse flow, and reattachment in a small region. A domain-average loss can underweight that region. The physics-guided zonal objective is

$$\mathcal{L}_{\text{zonal}} = \alpha\mathcal{L}_{U<0} + (1-\alpha)\mathcal{L}_{U\geq 0}, \tag{6}$$

where each regional mean is normalized by its own number of points.

FlowMLLab keeps five heights for development, H33/H58 for validation, and H44/H67 for the held-out classroom evaluation. The validation rule selects $\alpha = 0.6$. The article and classroom figures use equal x/H–y/H scaling; the measured data domain has $L/H \simeq 5$.

The pinned article-repository inference path constructs each local input patch from the same DSMC U, V field that it reconstructs. Its stored C-DeepONet panels are therefore target-assisted reconstruction evidence, not an independently deployable held-out map. A nearest-sample baseline on that patch gives 4.40% at H44 and 6.79% at H67, lower than the stored model's 4.92% and 8.23%. The classroom model instead receives only height, coordinates, and known geometry.

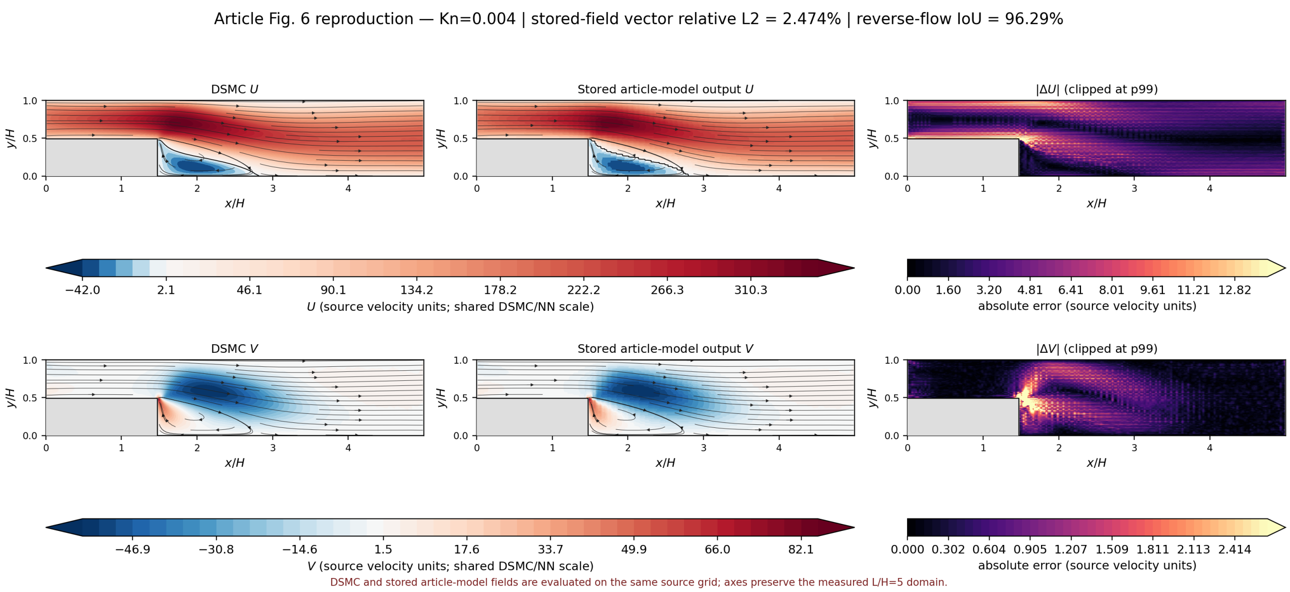


Figure 1: Micro-step rarefaction case at Kn = 0.004: DSMC, stored target-assisted article-model output, and absolute velocity error. The solid step remains masked and the field color limits are shared.

5 Micro-Step: Height Variation and Local Metrics

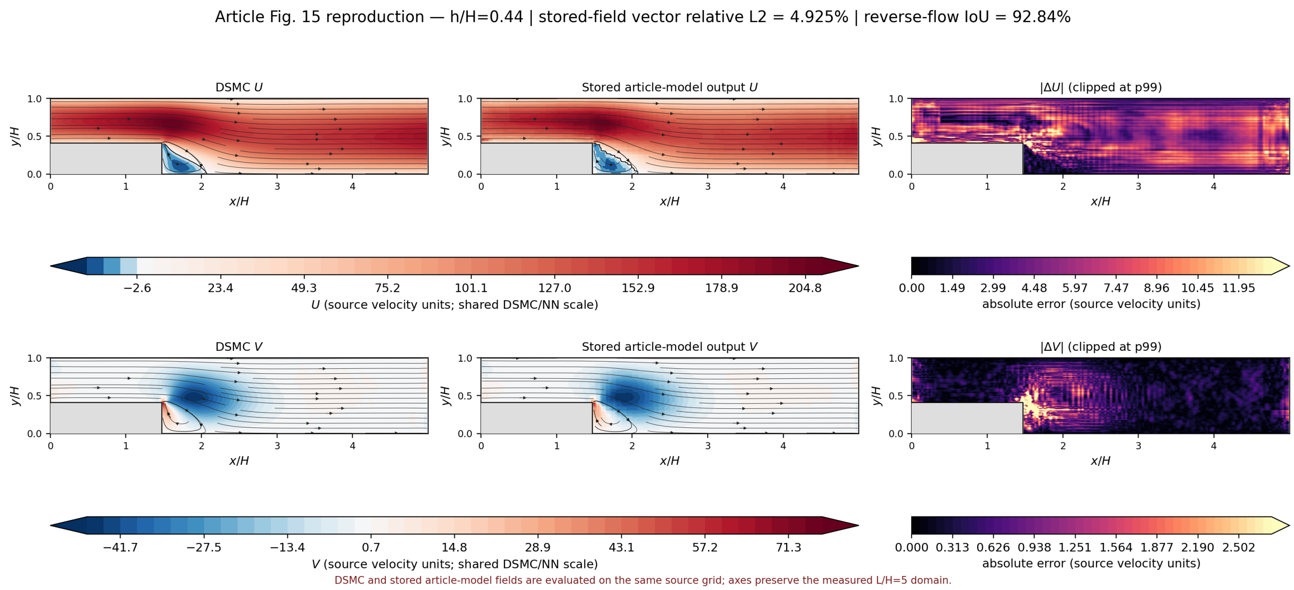


Figure 2: Article height-variation case H44, rebuilt on its source grid. Reverse-flow IoU and reattachment length complement the vector relative L_2 value.

The independently trained FlowMLLab classroom model provides a second comparison at H44/H67:

Case	Vector L_2	Reverse-zone L_2	Reverse-flow IoU	L_r/L (DSMC/model)
H44	9.594%	19.863%	93.27%	0.1199 / 0.1138
H67	11.267%	33.257%	80.49%	0.1042 / 0.1149

The local improvement is purchased with a substantial global tradeoff: unweighted-to-zonal error changes from 7.228% to 9.594% at H44 and from 5.934% to 11.267% at H67. The latter is almost a factor of two; describing it as a modest increase would contradict the held-out table.

Decision rule

The zonal model is accepted only if validation shows the planned reverse-flow improvement without violating the predeclared global-error allowance. The held-out geometries report the result; they do not select α.

6 Micro-Nozzle Back Pressure: New FlowMLLab Fields

The public nozzle dataset contains 15 complete DSMC snapshots on a common 101-by-31 curvilinear grid. FlowMLLab preserves density, temperature, U, V, Mach number, pressure, and Knudsen number. For each of the six article outputs, physical-coordinate and shock-aligned interpolation are compared on 12 development pressures. The aligned route detects the downstream density compression in each source-field row, interpolates its location in pressure, translates both source rows to that predicted surface, and only then blends them. It does not inspect the held-out target field.

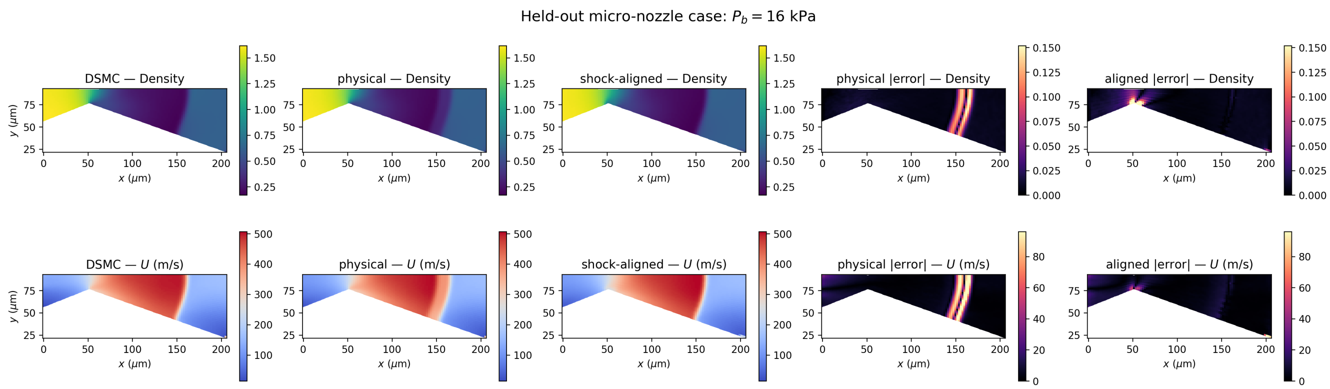


Figure 3: Held-out $P_b = 16$ kPa comparison of DSMC, physical-coordinate interpolation, and shock-aligned interpolation. Shared field and error scales expose how alignment removes most translated-shock smear in U while preserving the nozzle aspect.

The largest discrepancies concentrate around the translated compression/shock region. This explains why apparently good upstream fields can coexist with a larger global velocity or Mach error.

7 Back-Pressure Profiles and Error Matrix

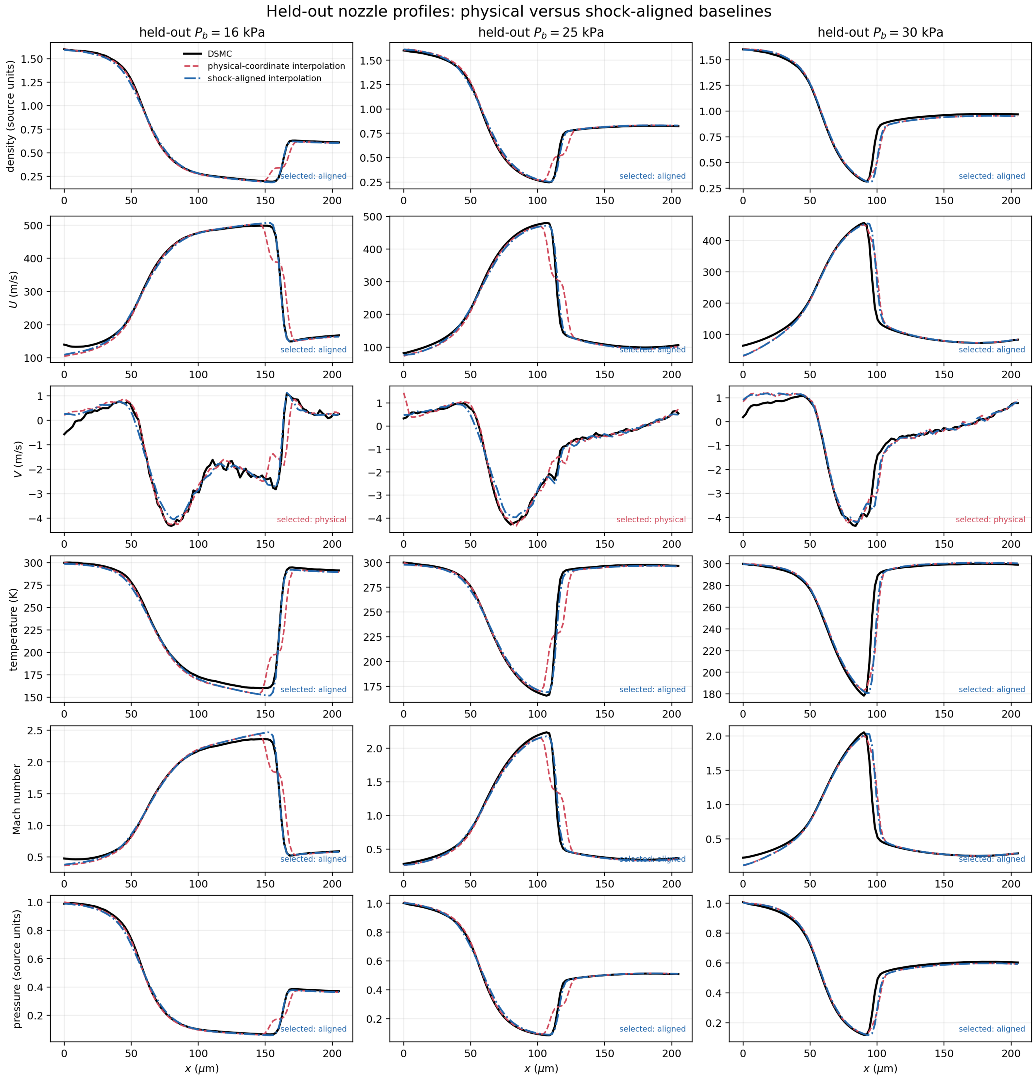


Figure 4: Centerline slices for both interpolation baselines at all three held-out back pressures. The selected route is frozen by development-only folds. These curves are regenerated by `qa/run_nozzle_field_validation.py`.

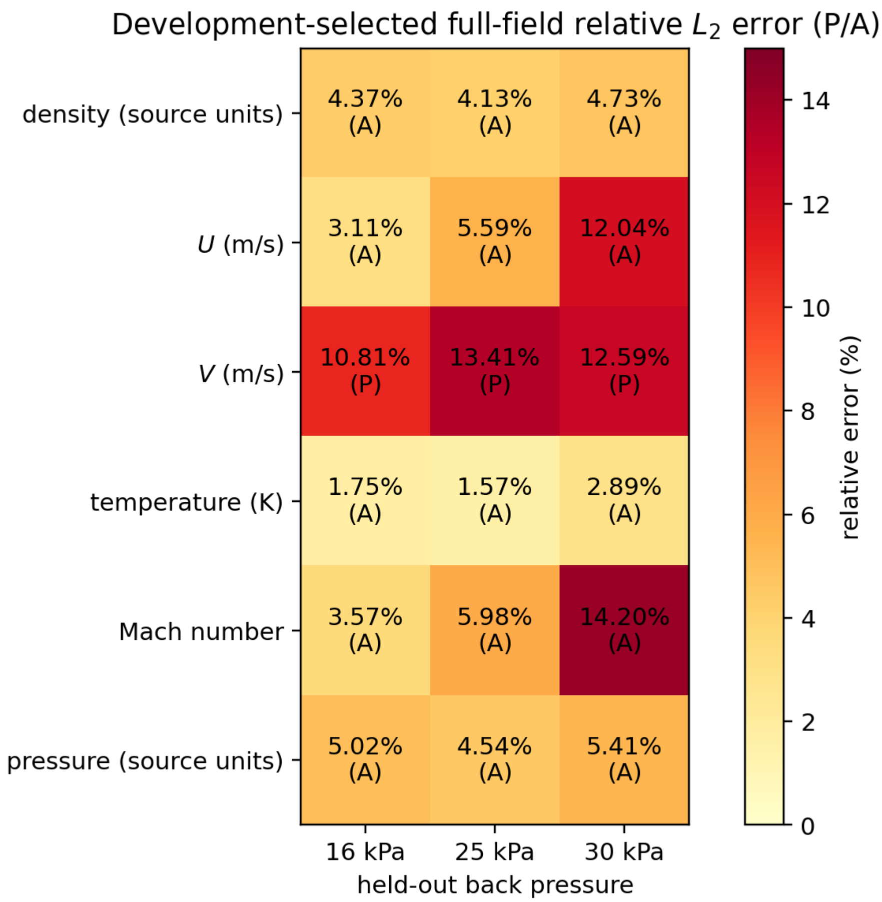


Figure 5: Development-selected full-field relative L_2 errors. Each cell marks physical (P) or aligned (A); a single average would hide output dependence and the harder 30 kPa case.

Field	16 kPa	25 kPa	30 kPa
Density	4.366%	4.130%	4.734%
U	3.107%	5.586%	12.038%
V	10.813%	13.409%	12.593%
Temperature	1.751%	1.572%	2.893%
Mach number	3.574%	5.984%	14.200%
Pressure	5.020%	4.539%	5.406%

8 Shock-Centered Coordinates

A moving sharp feature can require many linear modes in laboratory coordinates. For detected compression location x_s and width δ_j, define

$$\xi_j = \frac{x - x_s}{\delta_j}, \qquad \widetilde{\rho} = \frac{\rho - \rho_R}{\rho_L - \rho_R}. \tag{7}$$

Across all 15 DSMC centerlines, FlowMLLab reproduces the retained POD audit:

Coordinate	Mode 1 energy	Modes 1–2	Modes 1–3	Modes for 99%
Physical x	78.922%	88.467%	93.762%	8
Shock-centered ξ_j	97.986%	99.182%	99.865%	2

Alignment reduces representation rank; it does not by itself solve deployment. A new case does not arrive with x_s. The notebook therefore evaluates a pressure-to-shock-location component separately and retains the non-monotone 22 kPa detection as a diagnostic rather than silently deleting it.

Learning checkpoint

Ask two questions about every learned coordinate transform: (1) how is it estimated for a new input? (2) how does its uncertainty propagate into the reconstructed physical field?

9 Throat-Location Variation: Reproduction Boundary

The article also studies a geometry branch in which x_{throat}/L varies. The training locations are

$$0.10,\ 0.15,\ 0.20,\ 0.25,\ 0.35,\ 0.45,\ 0.55,$$

and the reported held-out location is 0.30. The article reports centerline errors of 4.07% for density, 9.78% for U, 10.85% for Mach number, and 4.82% for pressure.

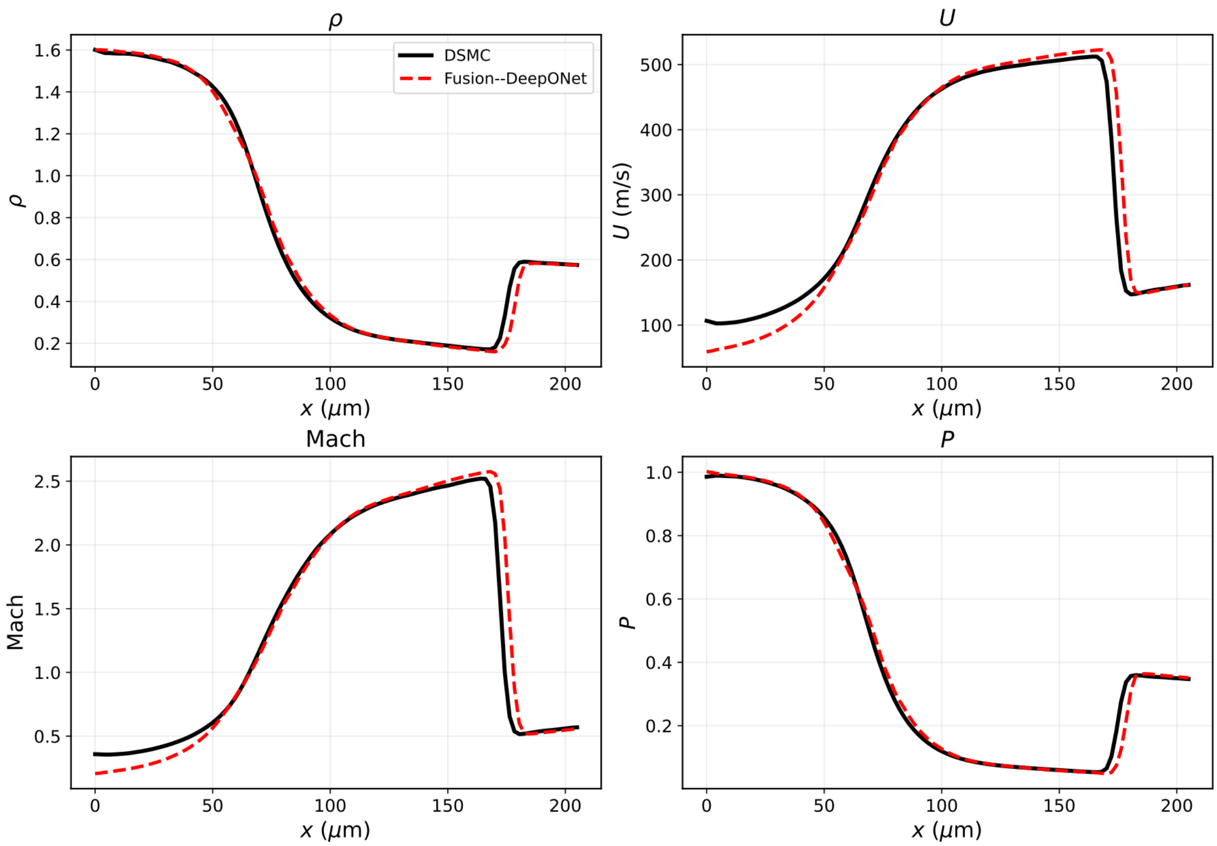


Figure 6: Published throat-location centerline comparison at $x_{\text{throat}}/L = 0.30$, reproduced here as a literature reference under CC BY 4.0. It is not a FlowMLLab-generated field. Source: Roohi and Mahdavi, 2026, arXiv:2605.12723, Fig. 20.

Scientific boundary

The public reproducibility repository presently contains the 15-case back-pressure DSMC sweep, but not the throat-geometry DSMC sweep or its trained checkpoint. Consequently, FlowMLLab regenerates the back-pressure result from code and treats the throat plot only as a cited reference. A full throat rerun requires publication of those geometry-indexed fields.

10 Laboratory Workflow and Assessment

10.1 Lab 1: micro-step

1. verify hashes, row counts, units, masks, and equal physical aspect;
2. inspect one DSMC height field and identify separation and reattachment;
3. fit the coordinate baseline on development geometries;
4. select α using H33/H58 only;
5. evaluate H44/H67 using global, reverse-zone, IoU, and L_r metrics; and
6. compare the independently trained contours with the retained article cases.

10.2 Lab 2: micro-nozzle

1. verify the 15 public DSMC files and their full-field derivative;
2. detect x_s, reproduce physical versus aligned density POD, and inspect 22 kPa;
3. freeze the 12/3 pressure split, two baselines, local metrics, and global guardrail;
4. run `qa/run_nozzle_field_validation.py`;
5. interpret every row of the 6-by-3 error matrix and the 16 kPa contour; and
6. propose the additional data package required for the throat study.

10.3 Recommended grading evidence

Category	Required evidence
DSMC verification plan	spatial/time/particle/sampling studies, wall model, conservation, uncertainty
Data integrity	source commit, hashes, units, coordinates, masks, case table
Selection discipline	development/validation/held-out cases and frozen rule
Field fidelity	full-field and local errors, topology or shock position, equal-aspect figures
Interpretation	one success, one failure, and one next numerical experiment
Reproducibility	restart-and-run notebook plus machine-readable metrics and figure manifest

11 Takeaways

1. DSMC labels must be verified as a particle calculation independently of any neural model.
2. Complete physical cases, not shuffled spatial points, define the experimental split.
3. Zonal metrics expose micro-step recirculation behavior hidden by a domain average.
4. Shock alignment compresses the nozzle snapshot family from eight to two density modes at 99% energy.
5. The selected interpolation baseline keeps every held-out full-field error below 14.21%, but this is baseline evidence, not a regenerated DeepONet claim.
6. A cited article plot and a code-regenerated result are different evidence and must remain labelled accordingly.

Decision rule

The next defensible software increment is a published throat-geometry DSMC archive with the same provenance, case metadata, full fields, and verification records as the pressure sweep. Only then should the throat operator be trained, selected, and evaluated in FlowMLLab.

Primary sources

- E. Roohi and A. Mahdavi, "Analysis of the rarefied flow at micro-step using a DeepONet surrogate model with a physics-guided zonal loss function," *Microfluidics and Nanofluidics* 30:44 (2026), doi:10.1007/s10404-026-02899-8.
- E. Roohi and A. Mahdavi, "Shock-Centered Low-Rank Structure and Neural-Operator Representation of Rarefied Micro-Nozzle Flows," *Physics of Fluids* 38, 082008 (2026), doi:10.1063/5.0343101.
- Public nozzle DSMC/POD package: github.com/Ehsan-Roohi/roohi-nozzle-pod-reproducibility, pinned in FlowMLLab at commit `e1b234b`.